

MOLLY MISHRA

DATA ANALYSIS • MACHINE LEARNING • SQL • AGENTIC AI

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Data-focused student with hands-on experience in Python, SQL, data visualization and machine learning. Built analytical dashboards, performed exploratory data analysis and developed AI/ML projects involving real-world datasets. Interested in applying data-driven analysis to solve business and operational problems.

SELECTED DATA ANALYTICS & VISUALIZATION PROJECTS

Customer Behavior Analysis | *Python • SQL • PowerBI*

- Analyzed customer behavior using Python, SQL and Power BI to identify purchasing patterns, customer segments and behavioral trends.
- Cleaned and transformed customer data using Python and SQL for exploratory and segment-level analysis.
- Built interactive Power BI visualizations to communicate key customer insights and support data-driven decision making.

Crimes Analysis | *Google BigQuery • SQL • Window Functions*

- Analyzed crime data using Google BigQuery to identify demographic, age-based and crime-category patterns.
- Applied CTEs, window functions, ranking and aggregation techniques to perform multi-dimensional analysis.
- Derived statistical insights from crime trends and demographic segments using advanced SQL queries.

Power BI Sales Dashboard | *DAX • Business Analytics • Data Visualization*

- Developed an interactive Power BI dashboard to analyze sales, profit, product performance and customer insights.
- Created KPI visualizations and interactive slicers to explore sales performance across different dimensions.
- Transformed raw sales data into business-focused visualizations for identifying trends and performance patterns.

AI AND MACHINE LEARNING PROJECTS

RED TEAM — AI-Powered Application Security Analysis | *TypeScript • AI Agents • Gemini • Application Security*

- Built an AI-powered red-team platform that analyses applications and codebases to identify real, evidence-backed security vulnerabilities rather than relying on generic vulnerability checklists.
- Designed an agentic analysis pipeline covering reconnaissance, attack-surface mapping, attack planning, security-test execution, evidence collection, validation and risk assessment.
- Implemented evidence-driven finding validation and deduplication using project fingerprinting and vulnerability fingerprints, reducing repeated findings across scans of the same application.

MNIST Neural Network — Built from Scratch | *Python • Numpy • Neural Network • Matplotlib*

- Built a fully connected neural network from scratch using NumPy to classify handwritten digits from the MNIST dataset, without relying on high-level deep-learning frameworks.
- Implemented forward propagation, loss calculation, backpropagation and gradient-descent-based weight updates to train the network.
- Developed the training pipeline including data preprocessing, weight initialization, activation functions and model evaluation to understand the underlying mechanics of neural-network learning.

CORE SKILLS

Programming: Python, SQL, Java, R,

Data Analysis: Pandas, NumPy, Data Cleaning, Exploratory Data Analysis, Statistical Analysis, Feature Engineering

Visualization: Power BI, Matplotlib, Seaborn, Excel

Machine Learning: Scikit-learn, Regression, Classification, Model Evaluation

Databases: MySQL / PostgreSQL / SQLite

Tools: Git, GitHub, Google Colab, VS Code

EDUCATION & RECOGNITION

BS in Data Science and Applications — Indian Institute of Technology, Madras | *Pursuing*

Hackathons & Recognition: Data Beyond Boundaries (IIM Bangalore) • Agentic AI hackathon (IIT Bhubaneswar) • Data Analytics Hackathon (IIT Madras) • Data Science Olympiad (SRMIST) • Ranked 100th out of 990 students in IITG data science competition